TODO - also check adherence against scheduled timees based on sleep diary not each other

TODO - log transform and adjust results. Include CIs

An Investigation into Time Agreement, Subject Adherence, and Patterns of Cortisol and DHEA in Novel Saliva Collection Device

Ezra Weible[1]

Introduction

0.5 pages; describe data and scientific hypotheses of interest. Rephrase scientific hypotheses into statistical hypotheses.

The data for this project was focused around at-home collection of saliva samples on strips of filter paper using a Saliva Procurement and Integrated Testing (SPIT) booklet to determine salivary cortisol and DHEA. The participants took samples four times a day for three days and recorded the time at which they collected their samples. The electronic monitoring cap on the collection plastic bottle also recorded the time. There were 31 healthy control

[1]Biostatistics Masters Student, Colorado School of Public Health, University of Colorado Anschutz Medical Campus, Aurora, CO

subjects testing this convenient and novel collection device. Subjects were asked to collect saliva samples at time of wake (identified by subject's reported wake time in their sleep diary), 30 minutes after, before lunch, and 600 min/10 hours after waking, though time of waking and lunch were allowed to vary by individual. There were three scientific research questions that this study aimed to answer: 1) What is the agreement between the subject's recordings of sampling times compared to the times recorded by an electronic monitoring cap?, 2) Are subjects adhering accurately to the +30min and +10 hour sampling times required by a study protocol?, 3) what are the changes of cortisol and DHEA over time? The purpose of this report is to develop the analysis plan to cover the following points based on these questions and report on the results:

1) Book time from wake (min) as predicted by cap time from wake (min) is in perfect agreement with an intercept of 0 and a slope of 1, accounting for subject-level random intercepts.

2) Descriptively, what proportion of subjects fall within $\pm$ 7.5 min (good adherence) and what proportion of subjects fall within $\pm$ 15 min (adequate adherence) at the 30min and 10hr measurements.

3) Is there a significant increase in cortisol and in DHEA from wake to 30 min? Descriptively, what is the rate of decline to the 10 hr measurement?

Methods

R v4.5.1 (2025-06-13 ucrt) was be used for all analyses. Initial data management was performed before analyses, including removal of DHEA measurements that were equal to 5.205 nmol/L (the upper detection limit) and removing subjects who had at least two DHEA measurements at this upper detection limit as there may be underlying conditions causing unusual values. Cortisol values greater than 80 nmol/L were also removed as they were clinically implausible. Booklet time, cap time, and sleep diary reported wake time were converted to POSIXct timestamps and differences between booklet time and wake time, cap time and wake time, and booklet time and cap time were stored in minutes.

For research question 1 (RQ1), a linear mixed model was used. The outcome was book time from wake, the primary predictor was cap time from wake, and a random intercept for subject was included. Using a 0.05 cutoff for p-values, the null hypotheses that the intercept is equal to 0 and the slope is equal to 1 were tested.

For RQ2, only descriptive statistics were used. The difference between the booklet and cap time for each sample were summarized by the proportion that fall within $\pm$ 7.5 min (good adherence) and what proportion of subjects fall within $\pm$ 15 min (adequate adherence) at the 30min and 10hr measurements.

For RQ3, a piecewise LMM with a knot at 30 min was used. The primary predictor was the difference between booklet time and wake time as it has fewer missing values than cap time and matches better with the sleep diary. The outcomes were cortisol or DHEA on the log scale to improve normality.

2 pages; past tense; no results; brief description of data management and cleaning. In depth description of analysis methods. Justify and explain data analysis approach.

Results

TODO - make table 1 and show missing percentages (data and missing data, descriptively differences between cap and booklet, description between cortisol and DHEA at each timepoint)

Number of participants removed: 1 Removed participant ID(s): 3037 Number of DHEA values at detection limit: 4

RQ1: Looking at the fixed effects, the t-values allow us to conclude that the intercept and slope are both significantly different from 0. Based on the residual for the random effects (1022.32), there appears to be large within-person variability. I also wanted to investigate to see if we fail to reject the null hypothesis that the slope is not equal to 1. There is no evidence that the slope differs significantly from 1 (p-value = 0.3889). Thus, changes in cap time do resemble the 1:1 line that the investigator described on the desired scatterplot. However, as the intercept differs significantly from 0, there is evidence of systematic bias, where the booklet time is 6.703 minutes earlier than the cap time, on average, at wake time. I plotted below the summary of these findings below. The red dashed line shows perfect agreement, and the blue line shows the modeled trend. Overall, the booklet and the cap agree well, but the booklet time is systematically earlier than the cap time, which is more obvious in the zoomed in section.

RQ2: Table

TODO - maybe make a figure to visualize this plot color coding

RQ3: Based on this model, the estimated cortisol level at wake time (0 min) is 7.222 nmol/L (intercept). The slope from 0-30min is 0.025 nmol/L/min, but it is not strongly significant ($t = 1.094$). This us unexpected because we were expecting to see a sharp rise in the first 30 min. Then, the slope after 30min is -0.009 nmol/L/min with strong statistical significance ($t = -8.101$). This can be visualized in the plot below.

TODO - adjust with CI and log transform

1-1.5 pages; present results from analysis. Also include descriptive statistics and model selection, if applicable. Reference tables and figures here. Full interpretation of results including pt estimates/CI/pvals/interpretation.

Discussion

0.5-1pg; interpret results in context of scientific Qs; discuss limitations

Limitations include having no ground truth, having a small sample size, assay limits, cap not always working,

Appendix

Directory: Github

File name: P0/Code/P0_Report.Rmd

Tables and Figures:

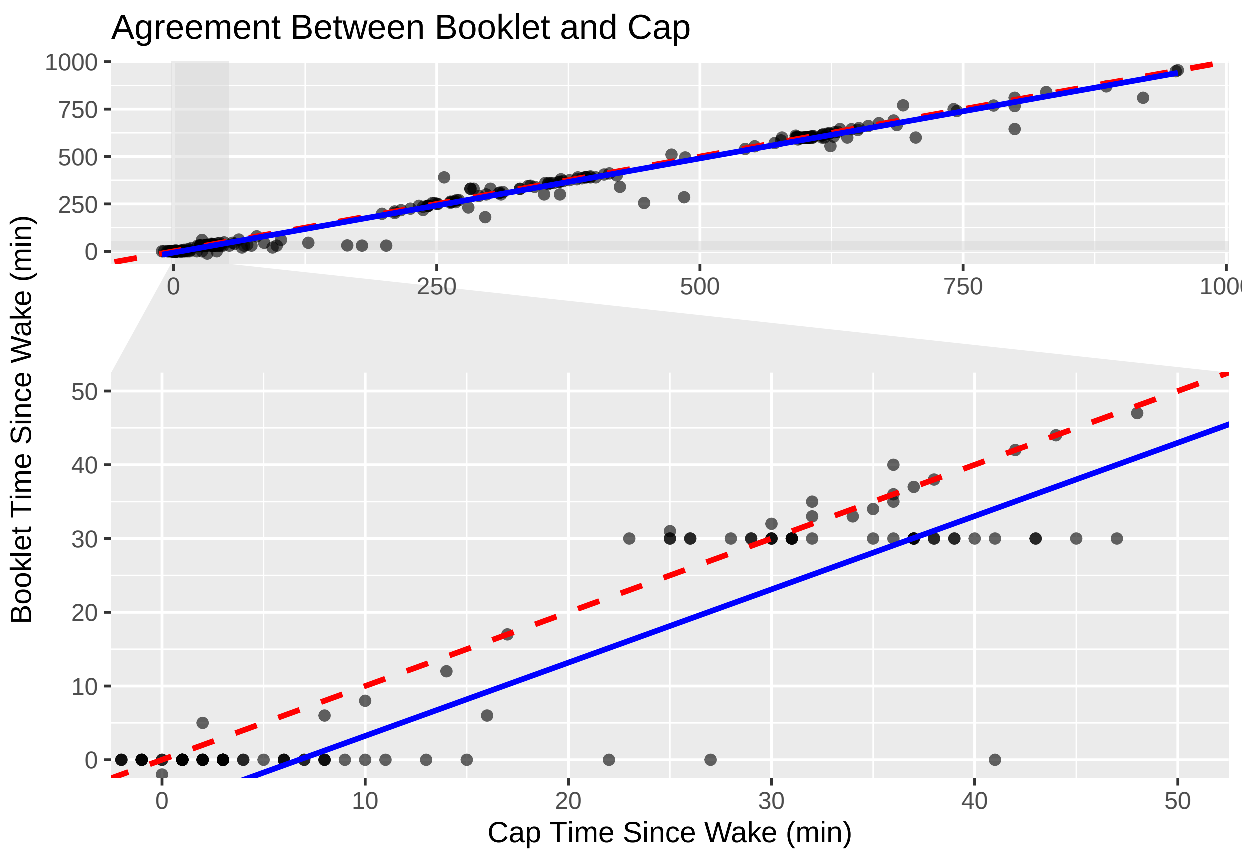

Figure 1: Research Question 1 Agreement between reported booklet time since wake and reported cap time since wake.

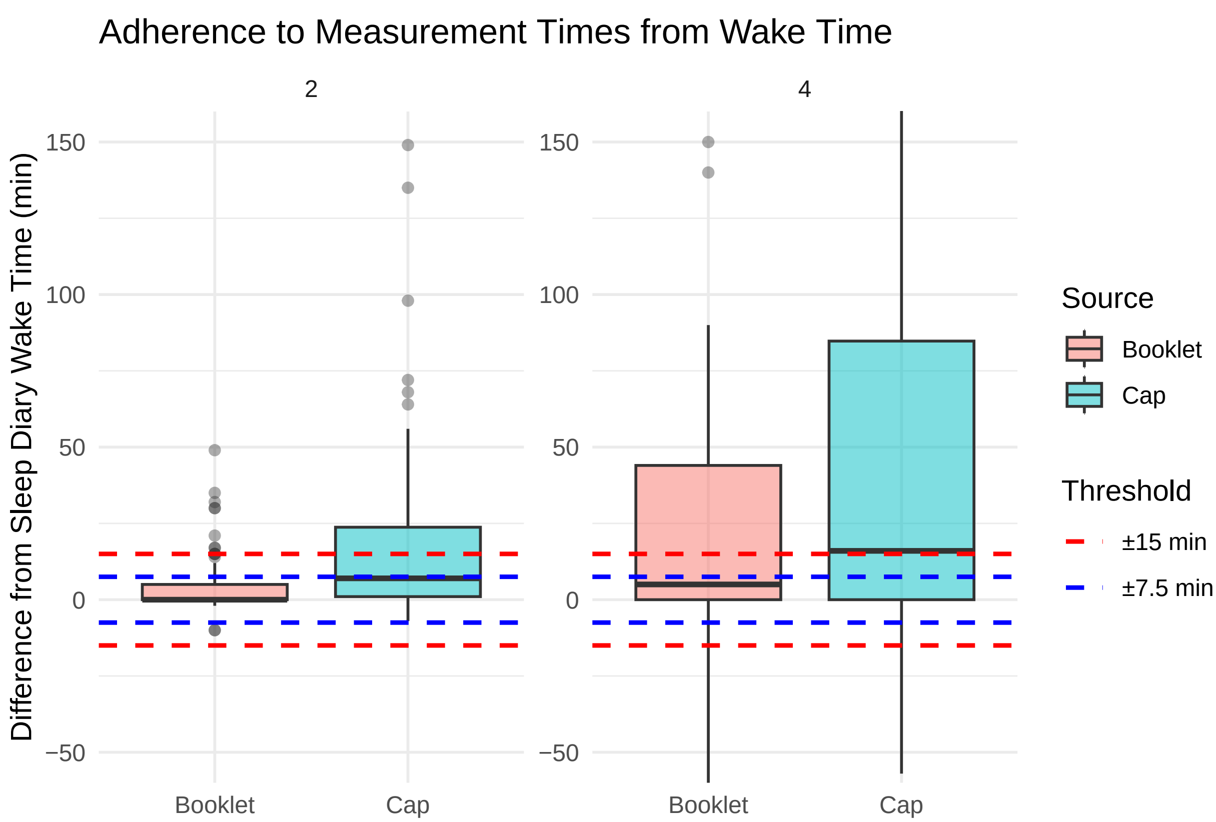

Figure 2: Research Question 2 Adherence to repoting windows.

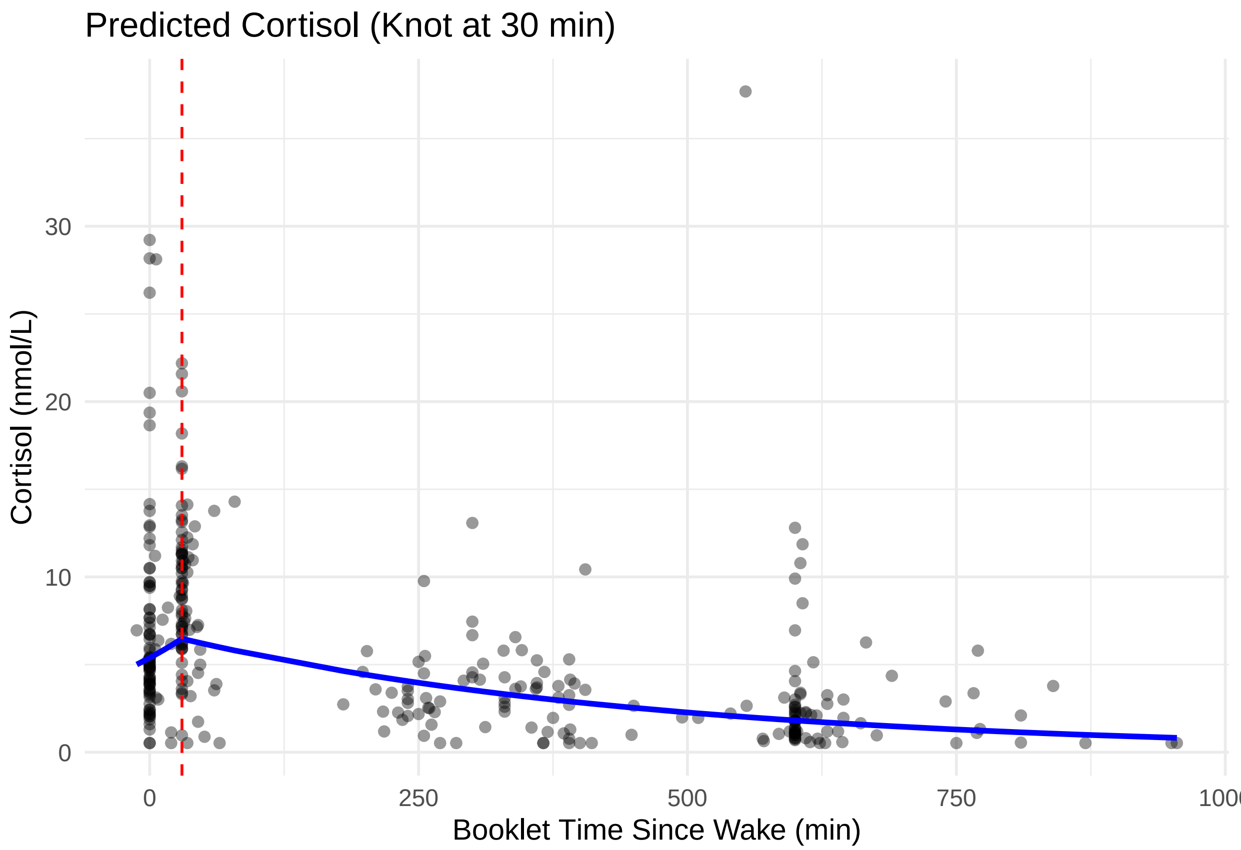

Figure 3: Research Question 3 Predicted trend of cortisol over time since wake, reported from booklet.

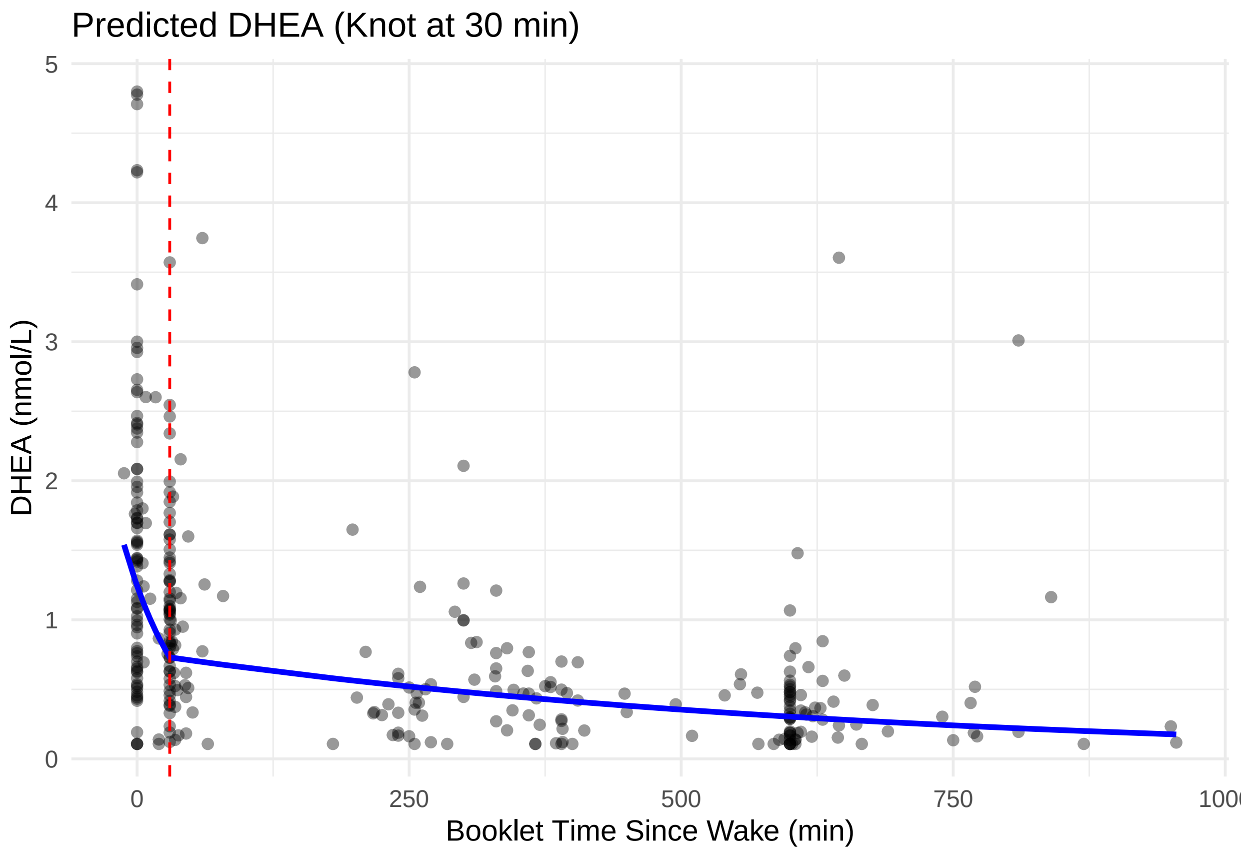

Figure 4: Research Question 3 Predicted trend of DHEA over time since wake, reported from booklet.

TODO - add subject level lines to graph